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EVALUATING PORE SPACE CONNECTIVITY BY NMR DIFFUSIVE COUPLING

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ABSTRACT

Nuclear magnetic resonance (NMR) measurements have been used as a way to evaluate reservoir rocks based, on the longitudinal and transverse relaxation times, T_1 and T_2 , and the time dependent diffusion coefficient, $D(t)$. Relaxation time distributions can be related to pore sizes and to the free and bound fluid fractions. In addition, there are well established equations correlating relaxation times to permeability. Regarding diffusion, at short times $D(t)$ is linked to the pore volume to surface area ratio while at long times it is related to pore tortuosity. Diffusive coupling between pores complicates the interpretation of NMR measurements because the relaxation time distribution will no longer be an accurate representation of the pore size distribution. On the other hand, analyzing the effects of diffusive coupling can provide information about connectivity within the pore space. In particular, knowing whether the macro-pores are connected to each other or are connected only through the micro-pores would provide valuable insight regarding the permeability.

In this work a low-field NMR spectrometer is employed to the NMR response of a micro-porous borosilicate glass bead pack. This material is a good representation of bimodal porosity systems. It consists of spherical grains with a mean diameter of 150 μm ; each of these grains is, itself, porous with an average micro-pore diameter of 0.3 μm . The system's total porosity is 80% and is divided equally between the micro- and macro-pores. Numerical simulations using three dimensional models incorporate the effects of diffusive coupling between micro- and macro-porosity and thus support the interpretation of our

measurements. Our experimental results show clear evidence of inter-pore diffusion and, therefore, add new information to conventional one-dimensional studies.

We have also applied this methodology to evaluate diffusion and connectivity in sedimentary rocks. Image analysis of thin sections provided the geometric parameters that were used as inputs to the computational models. Once again, we were able to simulate the NMR response and obtain good agreement with low-field NMR measurements.

INTRODUCTION

The use of NMR as a tool to evaluate porous media dates back to the early days of the technique. In 1956 Varian filed a patent describing a method to investigate groundwater in subsurface aquifers (Song, 2013). Later, Brownstein and Tarr (Brownstein, 1979) showed how magnetic resonance can probe pore size and shape information through diffusion and relaxation. Currently, NMR is one of the main logging tools used to evaluate reservoir rocks and is employed to estimate porosity, permeability and free and bound fluid content.

NMR measurements are sensitive to protons in the nuclei of hydrogen atoms in fluid saturated porous media. In the presence of an applied magnetic field, the ^1H spins are excited by a radio-frequency (rf) pulse. Once the excitation is over, the spins will return to their original state; this process is known as relaxation and is described by the longitudinal and transversal decay times – T_1 and T_2 , respectively.

Inversion-recovery and the Carr-Purcell-Meiboom-Gill (CPMG) pulse sequences are commonly employed to measure T_1 and T_2 , respectively (Levitt, 2002, Meiboom, 1958). The

fluid relaxation process in a porous material is written as

$$\frac{1}{T_{1,2}} = \frac{1}{T_{1,2,bulk}} + \frac{1}{T_{1,2,surface}}, \quad \text{Eq. 1}$$

where $T_{1,2,bulk}$ and $T_{1,2,surface}$ are, respectively, the bulk and surface contributions to the relaxation. If the effects of surface relaxation are relatively weak and if diffusion between pores can be neglected then the surface contribution to the lifetime can be written as

$$\frac{1}{T_{1,2,surface}} = \rho_{1,2} \cdot \frac{S}{V_p}, \quad \text{Eq. 2}$$

where $\rho_{1,2}$, is the surface relaxivity and S/V_p is the surface area to pore volume ratio. For brine contained in restricted geometries (e.g., the pores of reservoir rocks) surface relaxation plays a dominant role and the measured T_2 distribution can provide valuable information about the pore space.

In the simplest cases one might expect that each pore would contribute independently to the observed magnetization decay. The measured T_2 distribution would then be directly related to the distribution of pore sizes. However, experimental results show this is not always true. Because of diffusion between pores of different sizes, relaxation time distribution curves can be quite different than the underlying distribution of pore sizes (Ramakrishnan, 1999). Therefore, in order to estimate pore sizes from NMR responses, this phenomenon must be taken into consideration.

NUMERICAL SIMULATIONS

In order to simulate the model's NMR response random walkers are used to represent diffusing water molecules. Each walker carries a magnetic moment that can be relaxed (killed) when the walker hits the pore-grain interface. For larger pores or vugs the bulk fluid relaxation needs to be taken into account as well. As this mechanism depends only on the type of saturating fluid but not on the diffusion time, it can be included at the final step.

The method applied in this study makes use of variable step size random walks (Zhang et al, 2011). For dual porosity systems (such as reservoir rocks with large range of pore sizes) using a fixed

step size algorithm would be prohibitively inefficient because the step size would have to be small compared to the size of the throats in the smallest pores. In our variable step size algorithm, the step size is defined as the radius R of the largest sphere that can be drawn around the walker without intersecting the pore wall. The walker goes to a random point in the surface of this sphere and the walker's clock advances by $R^2/(6D_0)$, where D_0 is the saturating fluid diffusion coefficient. Once more, the radius of the largest sphere is determined and the walker steps randomly. The procedure is repeated until the walker reaches a distance δR to the pore wall, where the algorithm will change into a fixed step size walk. Once inside this shell of width δR , the walker will have two possibilities: stepping until it passes out of the shell and then returns to the variable step algorithm, or stepping until it hits the pore wall (inside the macro- or micro-pore space). When the walker hits the wall it has a probability, γ [$0 < \gamma < 1$], of dying (relaxing). γ is related to the surface relaxivity, $\rho_{1,2}$, and the diffusion coefficient, D_0 , according to

$$\gamma = \frac{2 \cdot \rho_{1,2} \cdot \varepsilon}{3 \cdot D_0}, \quad \text{Eq. 3}$$

where, ε is the step size when the walker is within the layer of width δR . The time at which the walker dies is recorded and another walker is released at a random location in the pore space.

MICRO-POROUS GLASS BEADS

Micro-porous glass beads are a good physical model for dual porosity systems (Figure 1), such as complex reservoir rocks. These are a commercial material used in liquid chromatography and are easily found in different grain sizes. We have selected borosilicate beads with a mean diameter of 150 μm . The system's total porosity is 80% which is equally divided between micro- and macro-porosity. According to the manufacturer's specifications, the system's pore volume to surface area ratio is $V_p/S = 0.31 \mu\text{m}$.

The packed micro-porous beads were saturated with water and experiments were carried out to measure the T_2 relaxation spectrum and time dependent diffusion coefficient, $D(t)$. Measurements were carried out in a Magritek Inc. Rock

Core Analyzer operating at 2.01 MHz. For the T_2 measurement, a CPMG experiment with 64 scans, 1024 echoes and 400 μ s echo spacing was performed. To measure $D(t)$ a pulsed-field gradient stimulated echo (Tanner, 1970) was employed. Gradient pulses were applied for 4 ms, the gradient strength was varied from 0 to 45 G/cm and diffusion times, t , ranged from 12 to 100 ms.

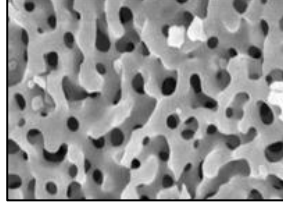


Fig. 1 Scanning electron microscope image of the interior of the micro-porous glass beads used in this study. (Millipore.com, 2014)

For the first tests on simulating the system's NMR response, the Micro-Grain Consolidation (μ -GC) model (Ramakrishnan, 1999, Zhang, 2011) was used. This model is based on a three-dimensional packing of overlapping spheres which are, in turn, comprised of micro-spheres (Figure 2). This allows us to represent mixed-porosity systems, such as rocks with bimodal pore size distributions.

The grains are arranged on a cubic lattice and by adjusting the radius of the macro and micro-spheres (i.e., the degree of overlap) one can control the macro- and micro-porosity of the system. The center to center distance can also be changed and is related to the over-all pore size. A scale factor controls the mismatch between the macro- and micro-grids.

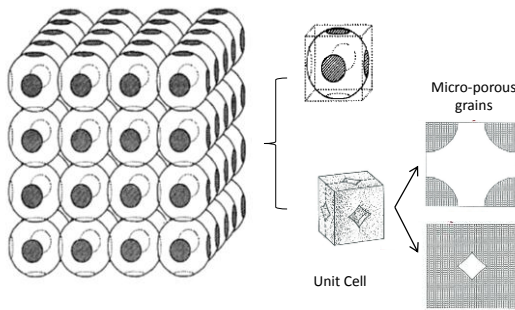


Fig. 2 Micro-Grain Consolidation model. On the left, the three-dimensional packing of macro-grains and, on the right, the unit cell and the micro-spheres contained within each macro-grain.

However, the simulation results did not match the measured $D(t)$ data. (Figure 3). From the scanning electron microscope image shown in Figure 1 (Millipore.com, 2014) it is clear that the micro-porous region is not well represented by overlapping grains. As an alternative, we proposed a Swiss Cheese geometry in which the micro-porous region is represented in terms of overlapping spherical pores (rather than grains). Calculations based in this model yielded much better results (Figure 3).

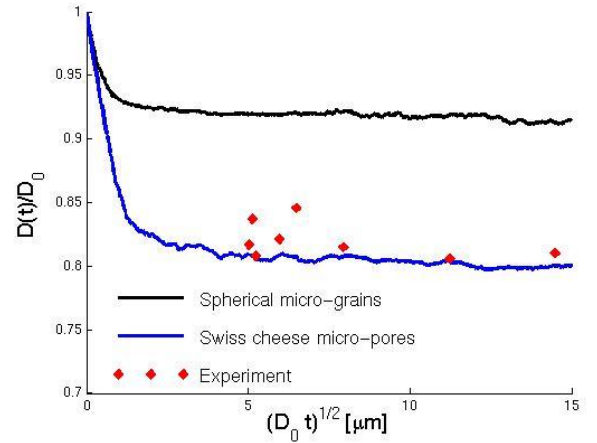


Fig. 3 Experimental and simulated diffusion coefficients (D) for different diffusion times (t). The red dots are the experimental data, with t ranging from 12 to 100 ms. The blue and black lines are the simulated data, the black line is the result from the μ -GC model (with spherical micro-grains) and the blue line is the result from the Swiss Cheese geometry (spherical micro-pores).

The Swiss Cheese geometry was also used to calculate the T_1 and T_2 lifetime distributions. In these calculations the values of ρ_1 and ρ_2 were adjusted to give the best fit and we see from Figure 4 that the comparison to the measured distributions is quite good. Both the experimental and simulated magnetization data were processed with a **Fast Laplace Inversion algorithm** (Song, 2002) to produce the T_2 distribution curves.

One feature of the distributions shown in Fig. 4 is quite important. Even though the micro- and macro-pores represent equal volumes, the T_2 distribution does not show two peaks with equal weight. Indeed, this is the signature of diffusion between the two sub-populations in the pore space

(Ramakrishnan, 1999). Note that for the T_1 distribution there is only a single broadened peak; looking only at this distribution one would not suspect the presence of two pore populations.

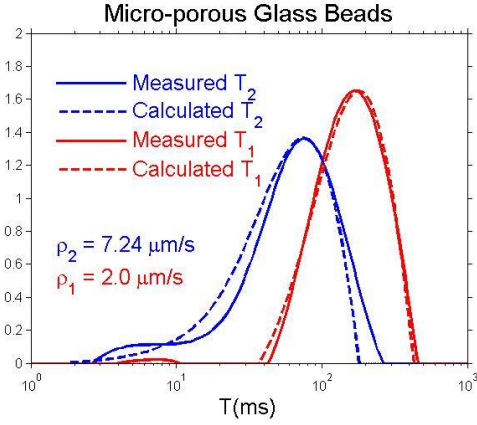


Fig. 4 Experimental (full lines) and calculated (dashed lines) T_2 (blue lines) and T_1 (red lines) spectrum for the micro-porous glass beads. A very good agreement between the data is observed.

In the next paragraphs, we describe the application of numerical simulations to the original μ -GC model (with spherical micro-grains) to sedimentary rocks.

IMAGE ANALYSIS

The geometrical parameters used as input to the μ -GC model are obtained through image analysis of thin sections. For each sample to be tested, a picture of the most representative area of the thin section is taken and processed. First, the visible macro-porosity (blue area) is counted and then micro-porosity is estimated by comparing this value to the gas porosity data. A binary image is created with the thin section pictures and grain and

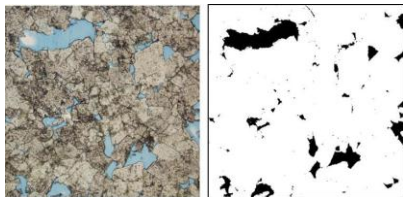


Fig. 5 On the left, the original thin section picture of the Silurian dolomite rock, where the blue area is the blue epoxy filled porosity. On the right, the binary image extracted used to measure pore sizes, with pores in black.

pore sizes and aspect ratios are measured. In Figure 5 it is shown the binary image for evaluating the pores extracted from the thin section for the Silurian sample.

ROCK SAMPLES

For the samples we have evaluated (Silurian Dolomite, Indiana Limestone, Red Massillon and Berea) we have tested only the μ -GC geometry with spherical micro-grains. The ρ_2 value used for the simulations were obtained through a Padé fitting of experimental D - T_2 maps, as described by Souza (Souza, 2013) and are listed in Table 1, together with porosity and permeability measured for the samples by Helium injection.

Table 1. Basic properties for the rock samples.

Sample	ρ_2 ($\mu\text{m/s}$)	Gas porosity (p.u.)	Gas permeability (mD)
Silurian	7	18.3	324.4
Indiana	8	15.1	157.1
Red Massillon	26	22.8	2,641.6
Berea	16	20.6	238

Before the NMR measurements, the samples were saturated with a 50,000 ppm KCl brine. In a Maran Ultra by Oxford Instruments (UK), operating at a Larmor frequency of 2 MHz for ^1H , CPMG experiments were carried out with an echo spacing time of 400 μs and the acquisition of 32 scans. The resulting decay curves were processed following the same procedure as for the glass beads.

In our numerical simulations the gas porosity and the surface relaxivity values, ρ_2 , were kept fixed for each sample. Given the value of ρ_2 we estimated the size of the micro-pores. The macro-pore dimensions and, if necessary, the weighting of micro- and macro-porosity were then varied to get the best agreement with the measured T_2 distributions. We found that the micro- and macro-porosities obtained from the thin section images could not be used directly in the simulations. This may be a consequence of the relatively small area covered by the images. The values of macro-porosity used in the simulations are compared to those derived from image analysis in Table 2.

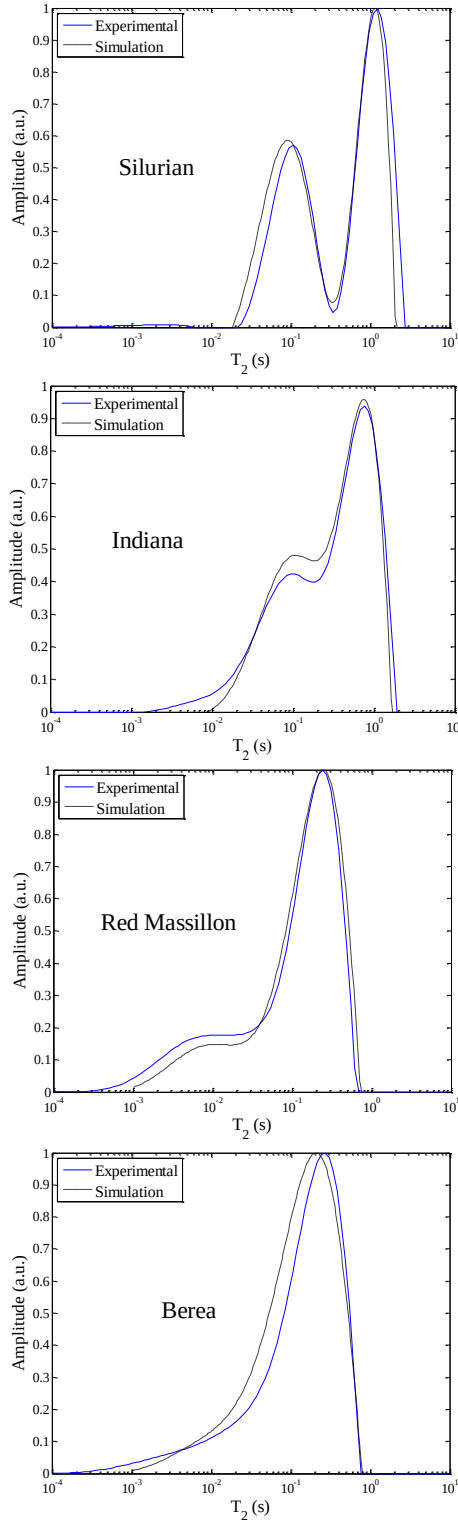


Fig. 6 Overlap of experimental and simulated T_2 distributions for the four rocks analyzed. The match between the measured and calculated data is quite satisfactory.

Table 2. Total porosity and macro-porosity fraction determined by Image Analysis and by the simulations for the four samples.

Sample	Total porosity (p.u.)	Macro-porosity fraction (p.u.)	
		Image Analysis	Simulation
Silurian	18.3	8.0	8.1
Indiana	15.1	3.4	7.5
Red Massillon	22.8	15.7	11.5
Berea	20.6	8.8	10.2

The magnetization decay curves obtained from the simulations were processed as before to get T_2 distributions. A comparison of the experimental and calculated distributions is shown in Figure 6. Generally the agreement is quite satisfactory. We note that in all cases the T_2 distributions have a distinct short time component representing the micro-porosity. However, comparing the area under the micro-porosity part of the distribution with the numbers presented in Table 2, we see that in all four cases there is clear evidence for coupling between the micro- and macro-pores. Indeed, the behavior of these rocks is qualitatively similar to micro-porous glass bead results shown in Fig. 4. In all four rocks the fraction of micro-porosity is roughly 50% but the area under the short time feature of the T_2 distributions is considerably smaller.

To further illustrate this point, we can repeat the simulations with a barrier that separates the micro- and macro-pores and eliminates diffusion between these populations. Results for the Indiana limestone are presented in In Fig. 7. If we measure to the left and right of the minimum in the barrier distribution, we see that the areas under the micro- and macro-peaks are now roughly equal. In addition, the position of the macro-peak has shifted to longer times. By contrast, the position of the micro-peak is essentially unchanged; most of the micro-pore relaxation occurs deep within the large micro-porous grain and is unaffected by the barrier. This shift implies that estimating pore sizes directly from the NMR T_2 distribution can be difficult because diffusive coupling shifts the position of the peak that represents the larger pores. This point is reinforced if we compare the T_2 distribution with MICP data (Fig. 8) for Indiana limestone. The MICP measurements

clearly indicate that, in the T_2 distribution, the large pores are shifted to smaller lengths. These considerations are especially relevant because it is the large pores that are most important in estimating permeability.

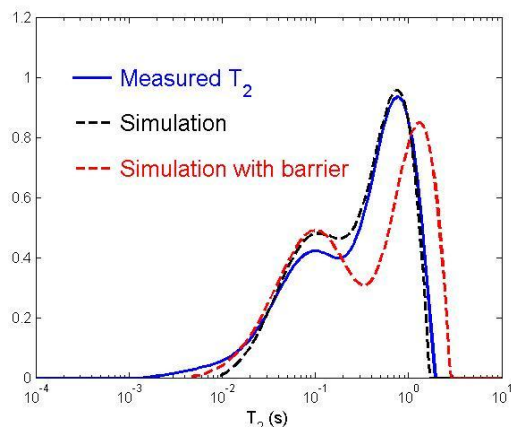


Fig. 7 For Indiana Limestone the measured T_2 distribution is compared to two simulations, one of which eliminate diffusive coupling.

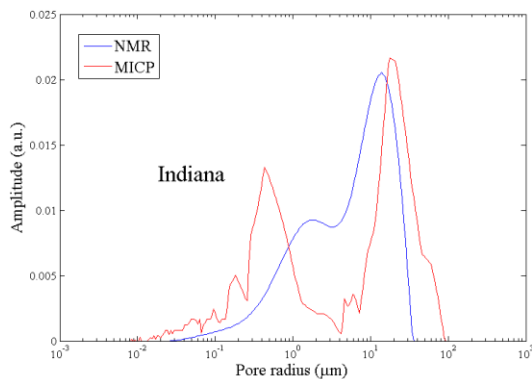


Fig. 8 For Indiana limestone we use ρ_2 to transform the time scale for the experimental T_2 distribution to a pore radius (radius= $2 \cdot \rho_2 \cdot T_2$) and then compare to MICP data (both areas are normalized to 1).

CONCLUSIONS

The μ -GC model together with random walk simulations is able to represent the essential features of the measured NMR properties of the different systems, both synthetic and natural, we have investigated here. Our results indicate that diffusive coupling (i.e., diffusion between large

and small pores) is important in all these systems. In subsequent work we will develop permeability models that take account of the linkage between micro- and macro-pores in these complex systems.

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